EVAA—Exchange Vanishing Adversarial Attack on LiDAR Point Clouds in Autonomous Vehicles

Chalavadi Vishnu, Jayesh Khandelwal, C. Krishna Mohan, and Cenkeramaddi Linga Reddy

***Abstract*— In addition to red-green-blue (RGB) camera sensors, light detection and ranging (LiDAR) plays an important role in autonomous vehicles (AVs) to perceive their surroundings. Deep neural networks (DNNs) are able to achieve cutting-edge 3-D object detection and segmentation performance using LiDAR point clouds. LiDAR-enabled AVs provide human perception by segmenting LiDAR point clouds into meaningful regions and providing semantic context to the AV user. However, the generation of point clouds to provide semantic segmentation in AVs is not reliable and secure, which may result in traffic accidents. We propose a novel adversarial attack against LiDAR point clouds in AVs in this article. We devised an exchange vanishing adversarial attack (EVAA) to deceive LiDAR point clouds by introducing targeted noise on specific objects (e.g., vehicles and driveways). In two autonomous driving datasets with 3-D object annotations, NuScenes and PandaSet, we evaluate the performance of our proposed attack framework. We achieve an attack success rate (ASR) of ≈63% and ASR of ≈29% on both NuScenes and PandaSet datasets, respectively.**

***Index Terms*— Adversarial attacks, autonomous driving systems (ADSs), light detection and ranging (LiDAR), point clouds, segmentation.**

I. Introduction

IN RECENT years, deep neural networks (DNNs) have become state-of-the-art methods for computer vision tasks, such as image classification [19], [22], object detection [29], [30], and image segmentation [8], [9]. This is mainly due to the ability of DNNs and light detection and rangings (LiDARs) to accurately model complex data. However, the effectiveness of such models depends heavily on their robustness to the attacks, which in turn target the model itself (adversarial learning attacks). Adversarial attacks have become a major focus of the machine learning research community, which has now evolved into its own field of study. These adversarial attacks are dangerous in autonomous driving systems (ADSs) because they mainly use LiDARs to navigate real-world traffic and have a direct impact on lives and property. DNNs grab the initial environment of the autonomous vehicle's (AV's) surroundings, which is followed by LiDAR sensors to finally verify the information. Therefore, the design of new adversarial attacks serves as a forewarning and verification mechanism for users of DNN models. In this article, we focus on the adversarial attack to vanish the original LiDAR point clouds as seen in Fig. 1 of various targeted regions from the perspective of AVs.

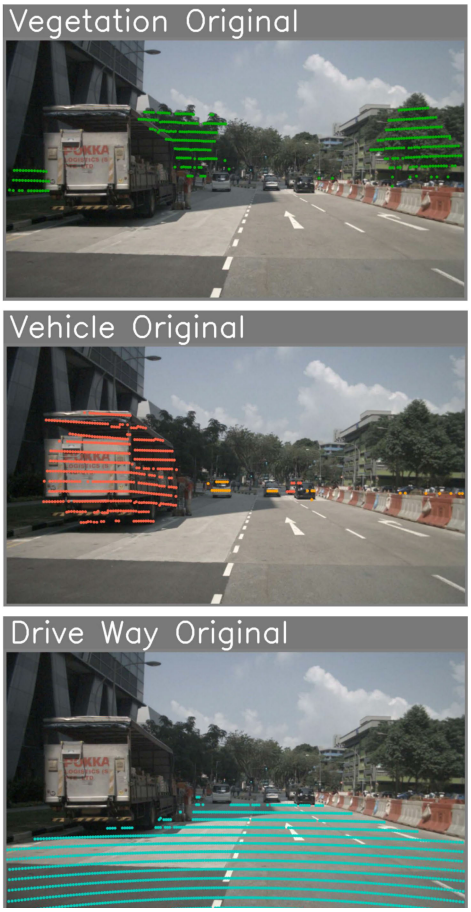


Fig. 1. Targeted vanishing on original LiDAR point clouds.

To the best of our knowledge, few research works [39] have attempted to implement a camera-based physical attack on vanishing a specific class where a person was chosen randomly in the MS-COCO dataset [23] to vanish his detections (invisibility cloak) on multiple state-of-the-art object detectors by introducing custom adversarial loss function to generate perturbed images. In another interesting work, namely the translucent patch attack [50], the author has a similar aim to hide instances of a targeted class by generating adversarial patches based on pixel information that is closer to physical objects in the real world. The evaluation of this method was

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done in an open-world scenario with no benchmark dataset as a reference. Another work closely relevant to our work and the basis for EVAA is targeted objectness gradient (TOG) attack [11] in which they perform multiple adversarial attacks such as object mislabeling, fabrication, and vanishing attack on MS-COCO [23] and PASCAL VOC datasets [13] with YOLOv3 [29] and SSD [24] used as object detectors.

In AVs, it is crucial to detect an object correctly, where even misclassification can lead to fatal accidents. To target such flaws in the existing LiDAR-based detection model, we propose an adversarial attack by generating adversarial LiDAR points in order to target any class. We employ a dynamic LiDAR points cloud generation function that considers the detected LiDAR point cloud during adversarial training.

In this article, we study the effectiveness of the targeted adversarial exchange attack on two autonomous driving datasets with 3-D object annotations, NuScenes [4] and PandaSet [40], with the aim of misclassifying or vanishing the LiDAR-sensed object. The main contributions of the proposed method can be summarized as follows.

1) We present a class-specific adversarial attack to vanish any targeted LiDAR point cloud from an autonomous perspective.
2) We introduce a custom adversarial LiDAR point generation function named exchange function to create multiscale adversarial samples with respect to the LiDAR point cloud of the detected object.
3) We quantify the attack success rate (ASR) and performance degradation of our attack on varying LiDAR points on two autonomous driving datasets, NuScenes and PandaSet, to show the applicability of EVAA in real-world scenarios.

II. Related Work

Previous works pertaining to adversarial attacks in the computer vision domain can be mainly categorized into two categories: 1) digital attacks and 2) physical attacks. Digital attacks focus on model architectures where the attacker either has access to the model configurations (known as a white box attack) or does not have access to the architecture (known as a black box attack). In the case of white box attacks, the attacker will freeze all the layers while introducing another adversarial layer that adds noise to the input samples while training [6], [14], [44]. Black box attacks are usually two-pronged attacks where adversarial perturbations are sent to the model through input images and an adversarial loss function is introduced at the model output stage to keep track of the model's training performance and make necessary changes. This is also known as model poisoning, recent works are [33], [35], [43]. Physical attacks are similar to black box attacks, but instead of digitally placing the adversarial patches, the perturbed sample is printed out and put on an object in a real-world environment to test an application's effectiveness. Usually, black box attacks are performed on models that detect multiple objects, which are similar to AVs.

The field of AV technology has led to a rise of physical attacks where adversarial patches in the form of stickers are placed on objects to fool AVs by malicious actors. But, it is not so easy to fool an object detector, as any decent detector will produce thousands of proposals (priors) for all objects which are internally controlled by a nonmaximal suppression (NMS) module. NMS controls the overlapping bounding boxes by choosing the best one, which is usually the box having the largest area with the highest prediction confidence while ignoring other proposals. So, if an adversarial attack focuses on a certain kind of prior to generate proposals, the NMS will automatically choose another prior during training and try to minimize the error of that prior. There are other detection algorithms using LiDAR where a 3-D point cloud is created of the surrounding objects. So, when the attacker tries to attack the point cloud, the NMS module ignores the attack by considering it as random white noise. In our proposed approach, we attack the LiDAR points generated, which is much harder to achieve than attacking the output of the last layer in a classifier. Our focus in this article is to come up with a robust adversarial attack on AVs that use object detectors, which are described in Sections III and IV.

A. Adversarial Attacks Against DNNs

Generally, the effective way to generate an adversarial example x^* is to maximize the cross-entropy loss function $J(x^*, y)$. Goodfellow et al. [16] proposed a fast gradient sign method (FGSM) in such a way to introduce perturbations based on the sign of the gradient. This method worked well for specific applications fine-tuned for it, but the transferability of that perturbation to other models was limited due to the technique being prone to ending up in a local maximum. Initially, Moosavi-Dezfooli et al. [25] proposed a universal perturbation method to fool any image using random noise. However, all of these studies digitally tampered with the input provided to the model [14], [21], [26]. Due to the lack of constraints and reproducibility of adversarial samples, these attacks do not naturally transfer well to real-world environments such as autonomous driving scenarios with varying illumination conditions and better object detectors to defend against these vulnerabilities [34], [45], [48]. Object detection and segmentation [3], [27] have come a long way in the advancement of computer vision. However, most of the models are trained and tested in ideal scenarios; but in reality, the scenarios are not ideal, especially in autonomous driving. A major challenge that arises is the dual priorities of the DNN model, i.e., object classification and localization [12]. In autonomous driving, it is equally important to know what objects are in our surroundings as well as how close the objects are to the reference [37], [47], [49]. Using LiDAR-based models, we can easily address this issue in a 3-D space, where we already know the spatial distance of every object from the reference point, and it is also easier to classify objects in 3-D geometry.

Similarly, another challenge, namely, occlusion and illumination, is easily solved while using LiDAR sensors. AVs with LiDAR sensors can detect the depth of an object at night with insufficient light. This may lead to spoofing attacks by manipulating the LiDAR signal [2], [32]. In cases of cluttered or

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textured backgrounds, object detection may predict incorrect boundaries when the background is similar to the object, while LiDAR is robust with a clear separation in background and foreground due to the distance from a reference point. Due to the above challenges, AVs use LiDAR sensors along with object detection as a verification module.

B. Adversarial Attacks on LiDAR Sensors

Some recent attacks on LiDAR sensors [5] have tried to generate spoofed LiDAR data. In LiDAR spoofing, the attacker simulates real echo pulses; this fake generated data can fool a model to provide an incorrect classification for the object detection task. The spoofing adversarial attack is also effective on sensors by adding a specific adversarial object with fixed dimensions on top of a vehicle [1], leading to incorrect object detection. Rather than detecting the boundaries of the vehicle as the model is supposed to, it will detect the adversarial object, and the vehicle will be undetected. Alternatively, we can remove some points, like in the proposed method "Object Removal Attacks (ORAs)" [17]. While creating a 3-D point cloud of an object, we only take points that are in the region of interest (RoI). Thus, there will be background points injected into the object, causing the model to fail. In one of the recent works [46], meshes were used to accurately target point clouds to avoid the limitation using RoI. This helped the attacker to focus only on the LiDAR point clouds instead of the boundary regions. Using boundary regions for adversarial noise will improve the attack performance if there are LiDAR point clouds of multiple classes, which was implemented in [18] by using generative adversarial attacks. Generative adversarial networks (GANs) also help in understanding the perturbation's distribution characteristics to constrain the magnitude of noise.

C. Adversarial Attacks against AVs

To overcome issues of transferability, authors have started proposing custom loss functions appended to the training stage of object detectors [28], [29] to fool AVs. This was a breakthrough in the adversarial attack field since penalties can now be backpropagated for any model during the training stage for any part of the architecture [10], [15], [31]. Recently, artificial generation of adversarial patches is gaining interest where authors would try to generate adversarial patches to be placed on images to fool the detectors [20], [38], [39], [42] with the help of loss function mentioned above. The closer a generated patch is to the targeted class or object visually, the better it can fool state-of-the-art object detectors. Similar to patch generation, we try to generate LiDAR points to place on or remove from the detected image.

Unlike the works mentioned above, the perturbations caused by attacks were usually focused on a single object [39] or a fixed patch [11] and none of them in autonomous driving scenarios, especially on LiDAR sensors with fixed point clouds. So, we propose an exchange vanishing adversarial attack (EVAA) with an adversarial logic function which tries to bridge the gap between adversarial attacks on LiDAR point clouds and AVs by utilizing both the LiDAR information and its respective red-green-blue (RGB) information for multiple classes.

III. Proposed Work

In this section, we present our proposed adversarial attack on LiDAR points to change targeted objects in monocular image scenes from AVs, as shown in Fig. 2. The proposed architecture focuses on noise generation, vanishing attack, and the exchange of different classes' LiDAR points, which are described in the following sections.

Point Cloud Detection using DNNs: Generally, a DNN-based object detector takes an input point cloud x of image resolution $(W \times H)$ and generates N number of points in LiDAR boundary region $B = \{b_1, b_2, \ldots, b_N\}$, where $b_i = (x_i, y_i, w_i, h_i, C_i, \hat{p}_i)$ representing an XY coordinate pair (x_i, y_i) with dimension (w_i, h_i) consisting of object confidence score $C_i \in [0, 1]$, and M class probability vector $\hat{p}_i = (\hat{p}_i{}^1, \hat{p}_i{}^2, \ldots, \hat{p}_i{}^M)$. The generated point clouds are filtered based on a user-defined confidence threshold t (usually 0.5). The corresponding point cloud model is generally trained by optimizing three loss functions simultaneously, viz., $\mathcal{L}_{\text{pointcloud}}$ (object localization), $\mathcal{L}_{\text{exist}}$ (object existence), and $\mathcal{L}_{\text{class}}$ (object class label) until convergence

$$\mathcal{L}(\tilde{x}, B, \mathcal{W}) = \mathcal{L}_{\text{pointcloud}}(\tilde{x}, B, \mathcal{W}) + \mathcal{L}_{\text{exist}}(\tilde{x}, B, \mathcal{W}) + \mathcal{L}_{\text{class}}(\tilde{x}, B, \mathcal{W}) \quad (1)$$

where $\tilde{x}$ is the training sample, B represents boundary regions, and $\mathcal{W}$ denotes model weights. The final weights are provided as input to our multiscale patch generation module to generate adversarial point clouds to fool the LiDAR sensor to vanish targeted classes from the scene output.

A. Exchange Attack Method

After obtaining the point clouds, class-wise noise is introduced through logic operations to vanish (nullify) or exchange LiDAR points of a different class.

1) Noise Addition: From Fig. 2, we can observe that the initial step of generating attack images is by adding noise to a single class image. This will generate noisy point clouds for each image in the dataset. We represent the noisy images of these classes with an "*," i.e., for the images belonging to the class vehicle, the corresponding noisy image will belong to the class vehicle* and so on. Thus, after the generation of noisy images, we will have a total of six image classes, namely—Vegetation, Vehicle, Driveway, Vegetation*, Vehicle*, and Driveway*. We will treat these classes independently, henceforth exchange attack between any of the two is feasible. Here, we added noise to the whole image, not just on the LiDAR point cloud. This is because the noise should be similar to LiDAR points. The background image (i.e., the part of the image where the class is not present) can be assumed as the actual image and the LiDAR points are noise with respect to various classes. Hence, the noise generated will follow this assumption and generate more LiDAR points all over the image.

While adding the noise, we make it target generated, i.e., we fixed the true label for the image regardless of the image class. Here, we are fixing the true label to "Vehicle." We choose the vehicle as when an AV is detecting objects through LiDAR points if it detects all the objects as vehicles it will

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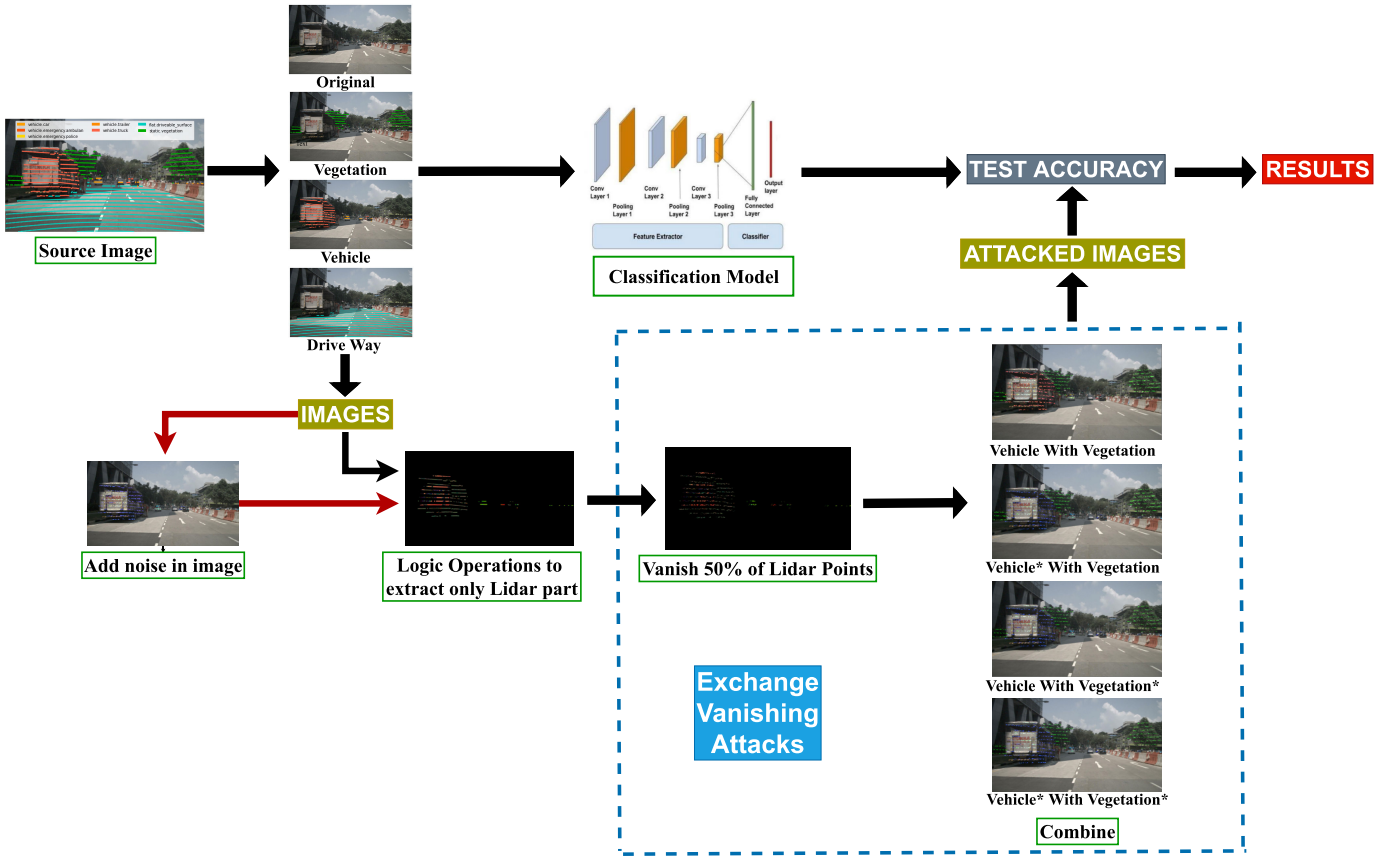


Fig. 2. Block diagram of the proposed method for EVAA where it is successful in making the targeted class "Vehicle" misclassified.

not move and hence not cause any mishappenings. This is safe as compared to detecting objects such as driveways where the vehicle can move which might lead to accidents. Thus, we add targeted noise to our adversarial attack methodology.

2) Logic Operations: Now, we extract the LiDAR points in a class image by using an XOR operator between the class image and its corresponding class-less image. With this operation, we obtain the LiDAR points of the non-noisy images. For noisy images, we have to further perform an "ADD" operator between the LiDAR point image of the corresponding non-noisy image and the noisy image. Finally, we will have the LiDAR points of all the original six classes. This step also ensures the noise added to the LiDAR point will be the only noise being considered an attack and the rest part of the image will remain unaffected.

3) Vanishing: To further our attack on images, we randomly vanish 50% of the LiDAR points from the logically extracted images. In Section V-C, we have experimentally shown the optimal percentage of vanished LiDAR points is 50%. This helps us to break any pattern a class might usually have. Thus making our model more vulnerable to classifying the images correctly and strengthening our attack.

B. Point Generation Function

For generating an adversarial sample x^*, a minor perturbation is added to the input image x after every epoch, with adversarial loss $\mathcal{L}^*$ and adversarial learning rate α^*. Though the difference between x and x^* is not perceivable to the human eye, the LiDAR detector is sensitive to the perturbations and fails to detect the object in the adversarial sample. Due to this, any LiDAR points detected in an adversarial sample will have the prediction confidence close to zero at the end of such training. The adversarial sample is generated iteratively using gradient ascent (since the generation of an adversarial sample is a maximization problem) as

$$x^*_t = \prod_x \left[x^*_{t-1} + \alpha^* \gamma \left(\frac{\partial \mathcal{L}^*(x^*_{t-1}, B, \mathcal{W})}{\partial x^*_{t-1}} \right) \right] \tag{2}$$

where $\prod$ is the projection of the adversarial sample onto a sphere centered for input image x to validate the pixel values, with γ being a sign function. Then, we retrain the model weights of the LiDAR detector and iteratively update the input image x along with the loss function $\mathcal{L}$ until the adversarial loss converges to a relative maxima. During training, if either of the defined termination conditions, i.e., maximal prediction loss or maximum iterations, is achieved, then the adversarial sample x^* at tth iteration (x^*_t) is finalized and used for the attack.

Combining: This is the final step of our attack, we will combine the vanished logically extracted images for different classes together to generate new attacked images. To achieve this, we perform an OR logical operation again. The OR operation on two different classes helps to combine them and form a completely new image/point cloud. The adversarial

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patches are generated by combining (1) and (2) as

$$x^*_t = \prod_x \left[x^*_{t-1} + \alpha^* \gamma \left(\frac{\partial P_{\text{loss}}(x^*_{t-1}, \phi, P_{\mathcal{W}})}{\partial x^*_{t-1}} \right) \right]. \quad (3)$$

Here, $\prod$ over x^*_t represents the point clouds with minimal adversarial loss $\mathcal{L}^*$ on targeted classes after introducing noisy LiDAR points. In summary, the proposed EVAA introduces the perturbations in the input image adapted to the point clouds detected by LiDARs.

IV. Experiments

Each of the six classes (Vegetation, Vehicle, Drive Way, Vegetation*, Vehicle*, and Drive Way*) is exchanged with the other four. For the Vegetation instance, we have exchanged it with Vehicle, Drive Way, Vehicle*, Drive Way*. We do not combine Vegetation and Vegetation* as it will just be a noisy-vanished attacked image of the same class. Thus we will have a total of 6*4 = 24 exchange combinations. We also will consider the three original classes to compare the attack percentage on our classes. Hence, we have a total of 27 test cases for every source image. For evaluation purposes, we use the testing dataset and generate all combinations of the attacked images. Furthermore, we predict the most probable class from a self-trained model and calculate the model accuracy for each combination.

A. Data Set

1) NuScenes: The nuScenes dataset [4] is a public large-scale collection containing 3-D object annotations for autonomous driving. We tag each LiDAR point from a keyframe in nuScenes with one of 32 potential semantic labels in nuScenes-lidarseg, which stands for LiDAR semantic segmentation. NuScenes-lidarseg now has 1.4 billion annotated points distributed among 40 000 point clouds and 1000 scenes (850 scenes for training and validation, and 150 scenes for testing). We have included nine background classes in addition to the 23 foreground classes (objects) from nuScenes.

2) PandaSet: The first academic and commercial dataset is the PandaSet [40]. PandaSet includes data from both a forward-facing LiDAR with image-like resolution (PandarGT) and a mechanical rotating LiDAR (Pandar) (Pandar64). A combination of cuboid and segmentation annotations was used to annotate the gathered data (scale 3-D sensor fusion segmentation). It includes 48 000 camera photographs, 16 000 LiDAR sweeps, and more than 100 8-s scenarios. There are a total of 28 annotation classes and 37 semantic segmentation labels in this dataset.

For this work, we have used three labels, namely, Vegetation, Vehicle, and Driveway, as they are common in both datasets.

B. Preprocessing

Initially, our proposed framework uses LiDAR point clouds as input, with each point belonging to three classes of either "Vehicle," "Driveway," or "Vegetation." The point clouds are then converted to RGB segmentation maps for the DNN to learn independent features of the targeted class and its background. We take a total of "n" images. For each of these images, we generate class-specific images. Here, three classes and one nonclass image are considered, so we will have $4 \times n$ images. The nonclass images are further used in the pipeline to generate LiDAR points by performing logic operations with respect to class images. The images are of shape 1395 × 784 (Nuscenes) and 1024 × 600 (PandaSet) and transformed into tensors of channel 3 RBG and (W × H) = (224 × 224). Fig. 3 shows the LiDAR point clouds and their respective class-wise RGB images that our proposed framework takes as input to the point-generation function. The possible combinations of images for various classes are shown in Table I.

C. Model Training

We require an image classification model to ensure our adversarial attack on the LiDAR images is successful as the class-specific accuracy decreases. We train a convolution neural network (CNN) consisting of conv-2-D layers and fully connected dense layers. We split the data into three components namely the training set, testing set, and validation set. We use the sparse categorical cross-entropy loss as our loss function in the model. The image tensor generated will be used as input, so, the output prediction will be an array of the probability of an image or set of images belonging to each class. For the classification of our testing dataset, we achieve an accuracy of more than 90.0% for the 3 classes, namely Vegetation, Vehicle, and Driveway, on both our datasets (NuScenes and PandaSet).

D. Transferability of Our Pipeline

Our method is a composition of various logical operations which are independent of any dataset or model. To transfer this to new data, first we need a classifier model (trained from scratch or pretrained) which will be used to calculate the prediction accuracy. Second, we just have to perform the logical operations in step-wise order to generate adversarial examples. Finally, we can test our results using the classifier model. To claim it, we presented results on two different datasets; NuScenes and PandaSet. Thus, we can say our method is easily transferable.

V. Results and Analysis

In this section, we evaluate our proposed exchange adversarial attack on the Nu-Scene and PandaSet datasets. We also present both quantitative and qualitative analyses of our proposed attack. We show the effectiveness of our proposed attack's success rate on two defense methods.

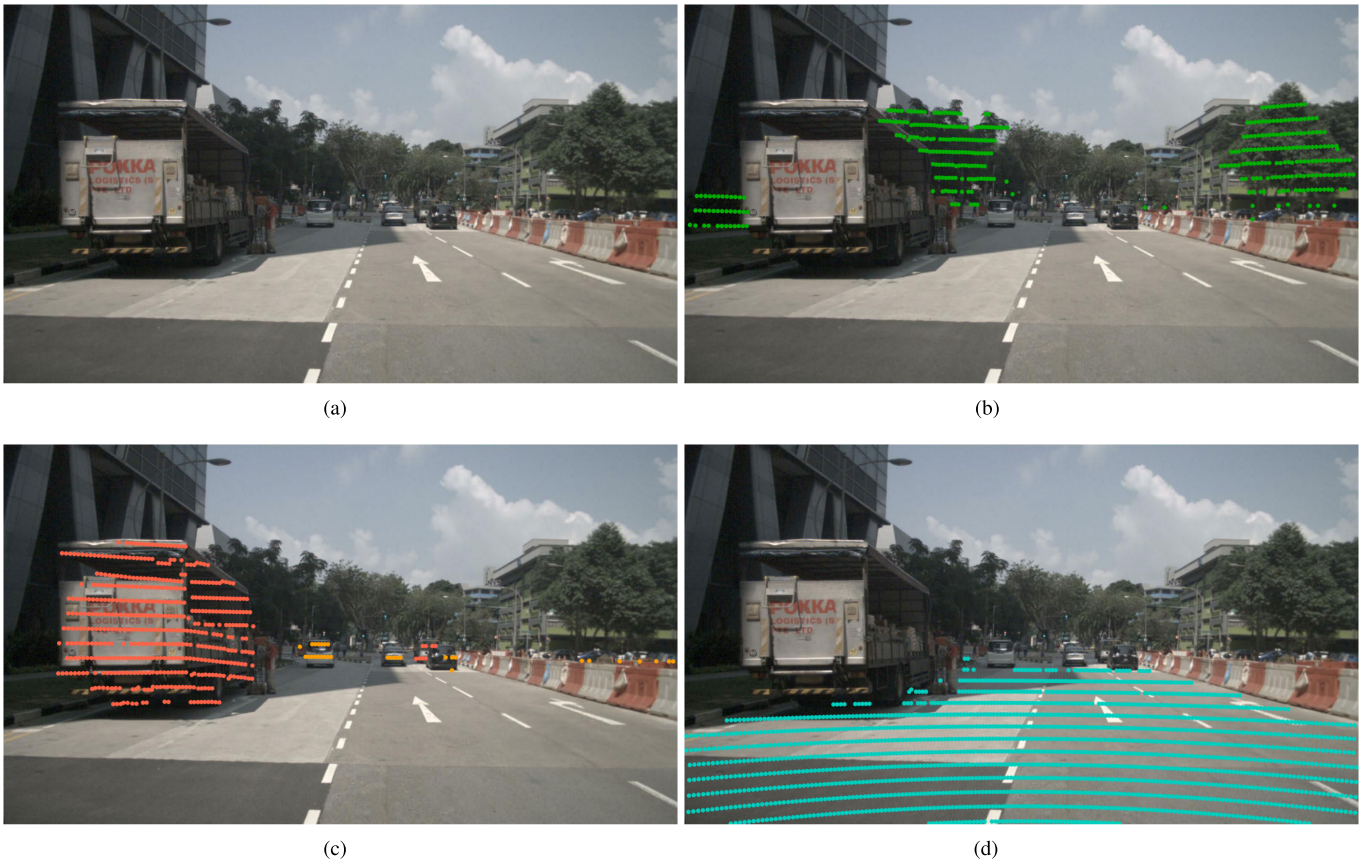
A. Quantitative Results

We evaluated our model on the generated attacked images using the exchange methodology. Attack Accuracy can be defined as the reduction in the classification accuracy of an exchange class with respect to the classification accuracy of the original class. From the exchange class accuracy of Tables II and III, it can be observed that our exchange methodology attack on LiDAR points is quite successful. Also,

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TABLE I
ALL THE POSSIBLE COMBINATIONS OF EXCHANGE IMAGES

Image Class	Vegetation	Vehicle	Drive Way	Vegetation*	Vehicle*	Drive Way*
Vegetation	✓	✓	✓	-	✓	✓
Vehicle	✓	✓	✓	✓	-	✓
Drive Way	✓	✓	✓	✓	✓	-
Vegetation*	-	✓	✓	-	✓	✓
Vehicle*	✓	-	✓	✓	-	✓
Drive Way*	✓	✓	-	✓	✓	-



(a) (b) (c) (d)

Fig. 3. LiDAR point clouds with their RGB Images. (a) Original image. (b) Image with vegetation LiDAR points. (c) Image with vehicle LiDAR points. (d) Image with drive way LiDAR points.

TABLE II
CLASS-WISE ATTACK ACCURACY (%) ON NUSCENES DATASET

Image Class original → Image Class exchanged ↓	Vegetation	Vehicle	Drive Way	Vegetation*	Vehicle*	Drive Way*
Vegetation	78%	34%	**60%**	-	22%	**16%**
Vehicle	66%	100%	92%	54%	-	40%
Drive Way	**10%**	**8%**	100%	**0%**	**8%**	-
Vegetation*	-	46%	64%	-	38%	18%
Vehicle*	78%	-	92%	62%	-	38%
Drive Way*	12%	60%	-	**0%**	62%	-

we have highlighted the lowest classification accuracy for "Vegetation," "Vehicle," and "Driveway" classes on Nuscenes and PandaSet datasets. In Table IV, the exchange class names are written in abbreviations as "Veg": Vegetation, "Veh": Vehicle, "Dw": Drive Way, "_": Exchange with, "*": Noisy Image.

1) NuScenes Dataset: From the Table IV we can observe that the Vegetation class's best attack was when we exchange noisy Vegetation image with Drive Way with or without noise addition on Drive Way. It has an **attack accuracy of 100%**, i.e., Vegetation* with Drive Way and Vegetation* with Drive Way* has a correct class prediction accuracy of 0%. Vegetation exchanged with vehicles (with or without noise) also has a significant attack as we can observe a significant reduction in classification accuracy for these cases. Similarly, for Vehicle class the best attack was when we exchange noisy or non-noisy

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TABLE III
CLASS-WISE ATTACK ACCURACY (%) ON PANDASET DATASET

Image Class original → Image Class exchanged ↓	Vegetation	Vehicle	Drive Way	Vegetation*	Vehicle*	Drive Way*
Vegetation	100%	66.67%	100%	-	5.13%	**0%**
Vehicle	33%	100%	**87.2%**	30.8%	-	**0%**
Drive Way	**0%**	**12.8%**	100%	**0%**	**0%**	**-**
Vegetation*	-	69.2%	100%	-	5.13%	**0%**
Vehicle*	94.9%	-	100%	94.9%	-	2.56%
Drive Way*	74.4%	100%	-	66.67%	97.4%	-

TABLE IV
BEST ATTACK ACCURACY (%)

Original Class	NuScenes Image	NuScenes Attack Acc	PandaSet Image	PandaSet Attack Acc
Vegetation	Veg*_Dw Veg*_Dw*	100%	Veg_Dw Veg*_Dw	100%
Vehicle	Veh_Dw Veh*_Dw	92%	Veh*_Dw	100%
Drive Way	Dw*_Veg	84%	Dw*_Veh Dw*_Veg Dw*_Veg*	100%

TABLE V
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART APPROACHES (%)

Methods	NuScenes	PandaSet
FGSM [41]	53%	65%
JSMA [36]	59%	71%
C&W [7]	68%	77%
GAN [18]	79%	82%
EVAA (Proposed)	**92%**	**98%**

Vehicle images with Drive Way. It has an **attack accuracy of 92%**, i.e., Vehicle with Drive Way* and Vehicle* with Drive Way* has a correct class prediction accuracy of only 8%. Vehicles exchanged with Vegetation (with or without noise) has a significant attack as we can observe a significant reduction in classification accuracy for these cases. Similarly, for the Drive Way class, the best attack was when we exchange the noisy Drive Way image with Vegetation. It has an **attack accuracy of 84%**, i.e., Drive Way* with Vegetation has a correct class prediction accuracy of only 16%. Drive Way exchange with Vehicle (with or without noise) also has a successful attack but not that significant as compared to Drive Way*.

2) PandaSet Dataset: From the Table IV we can observe that the Vegetation class's best attack was when we exchange noisy or non-noisy Vehicle image with Drive Way. It has an **attack accuracy of 100%**, i.e., Vegetation with Drive Way and Vegetation* with Drive Way have a correct class prediction accuracy of 0%. Vegetation (with or without noise) exchanged with a Vehicle without noise also has a significant attack as we can observe a significant reduction in classification accuracy for these cases. Similarly, for the Vehicle class, the best attack was when we exchange noisy images with Drive Way. It has an **attack accuracy of 100%**, i.e., Vehicle* with Drive Way has a correct class prediction accuracy of 0%. A Vehicle with noise exchanged with Vegetation (with or without noise) has a significant attack as we can observe a significant reduction in classification accuracy for these cases. Similarly, for the Drive Way class, the best attack was when we exchange noisy Drive Way image with Vegetation with noise, Vegetation without noise, and Vehicle without noise. It has an **attack accuracy of 100%**, i.e., Drive Way* with Vegetation, Drive Way* with Vegetation*, and Drive Way* with Vehicle have a correct class prediction accuracy of 0%. Drive Way with noise has a really successful attack, but Drive Way without noise does not give such good attacks.

Furthermore, we compared our proposed approach with various adversarial generation techniques to add noise to the LiDAR point clouds with respect to ASR. It can be observed from Table V that one of the basic attacks in FGSM is slightly better than random noise attack of 50%. But, FGSM did not perform well compared to flat perturbations introduced by Jacobian-based saliency map attack (JSMA) and gradient-based optimization approach (carlini and wagner). Also, these methods are not able to transfer well from fixed-point clouds. To overcome this limitation of dynamic point clouds, GANs are used to introduce targeted noise for different objective functions with respect to discriminator. Finally, the proposed approach was able to introduce noise through the point generation function based on boundary regions of LiDAR point clouds to outperform GANs.

Due to the targeted noise in our proposed vanishing adversarial attack, we can infer that the exchange attack takes semantic context into account. We can also observe that noisy images have more attack accuracy than original images as we predicted in our hypothesis. We can conclude that the attack with Drive Way is the most successful as it has a larger boundary region it will disrupt the LiDAR point cloud of the original class more than compared to other classes. However, other classes do not have the same impact on Drive Way due to their smaller boundary regions.

B. Qualitative Results

Fig. 4 shows the boundary regions of specific classes under various attack conditions. The *OriginalxOriginal* shows the predefined regions and their colors of Vehicle, Vegetation, and Drive Way. Here, we can observe "Drive Way" class has the largest boundary region compared to "Vehicle" and "Vegetation" classes. So when we apply the exchange attack on the original point clouds in *OriginalxAttacked* and *AttackedxOriginal*, "Drive Way" class is the most affected one as observed in Figs. 5–7. This also supports our quantitative

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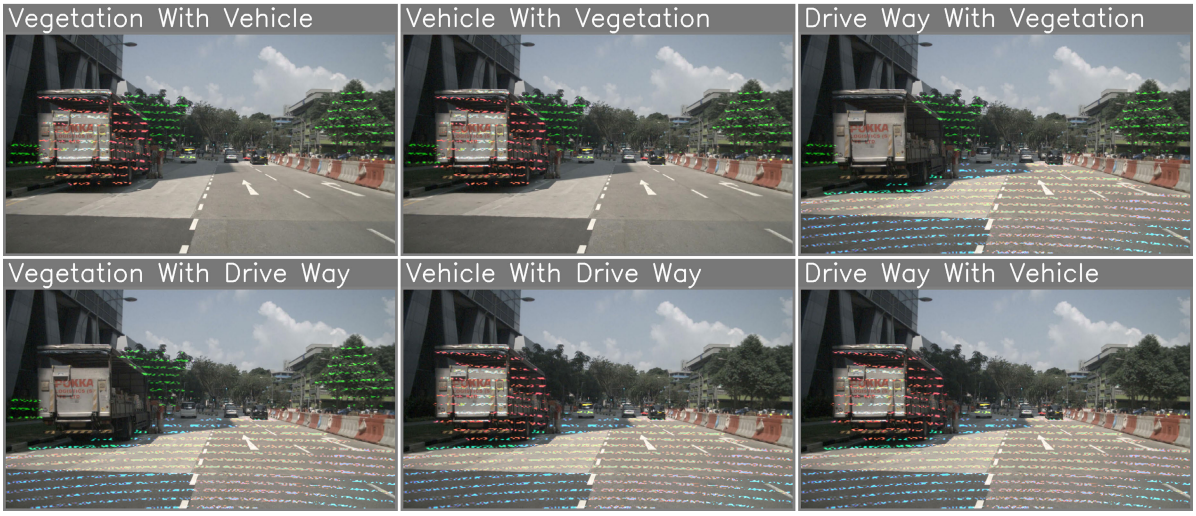


Fig. 4. Original X original.

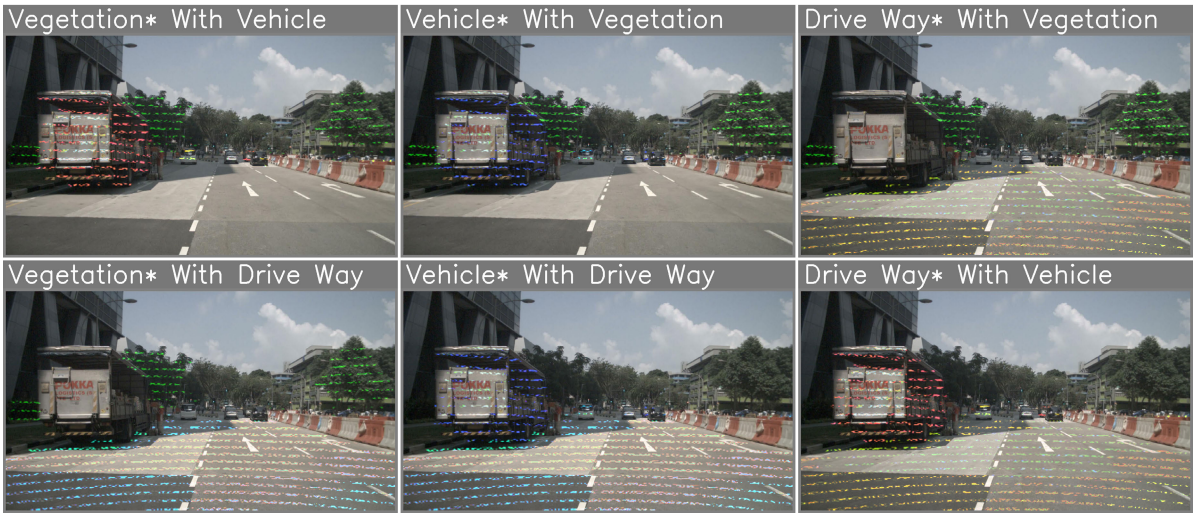


Fig. 5. Original X attacked.

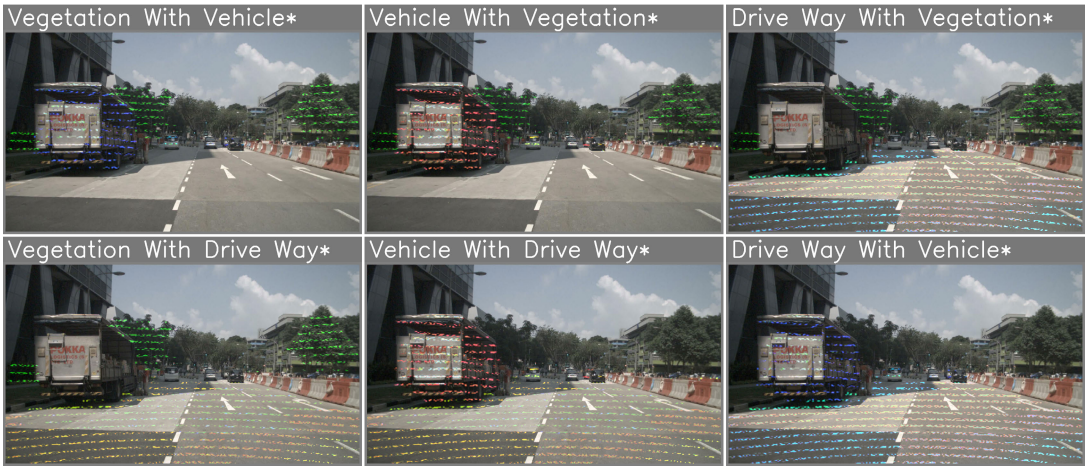


Fig. 6. Attacked X original.

analysis explained in the previous section. Although, "Drive Way" class ASR is highest in both datasets, Vegetation class's interaction with "Vehicle" class is similar in the PandaSet dataset. This is because the "Vehicle" and Vegetation classes have rich background information compared to "Drive Way" class foreground information. So, the targeted noise on the point clouds improves with better semantic information that we can extract.

C. Sensitivity to Vanish Points

One important hyperparameter (sensitivity) of our pipeline is the percentage of LiDAR points that should vanish as

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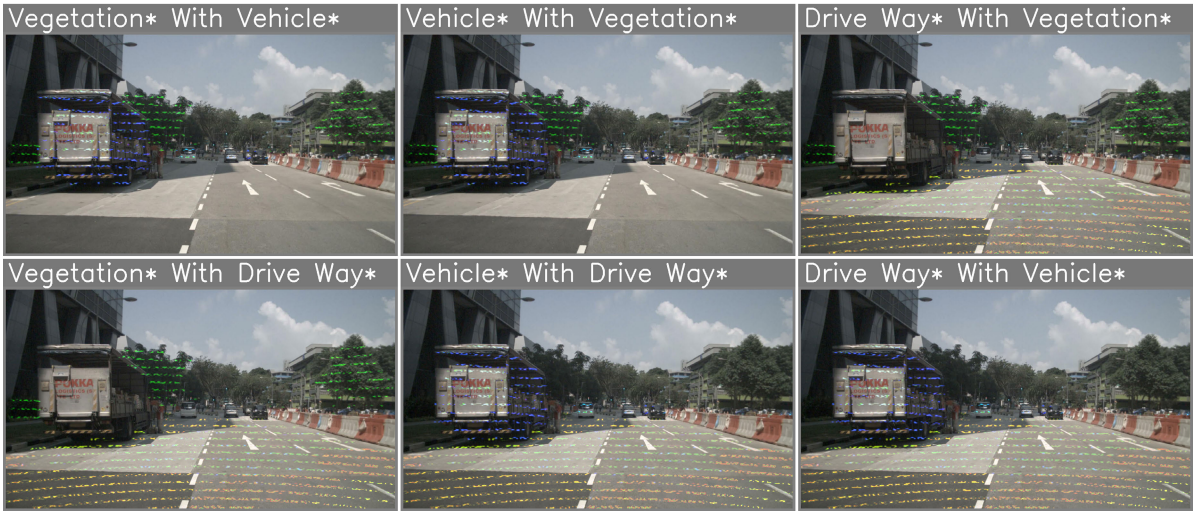


Fig. 7. Attacked X attacked.

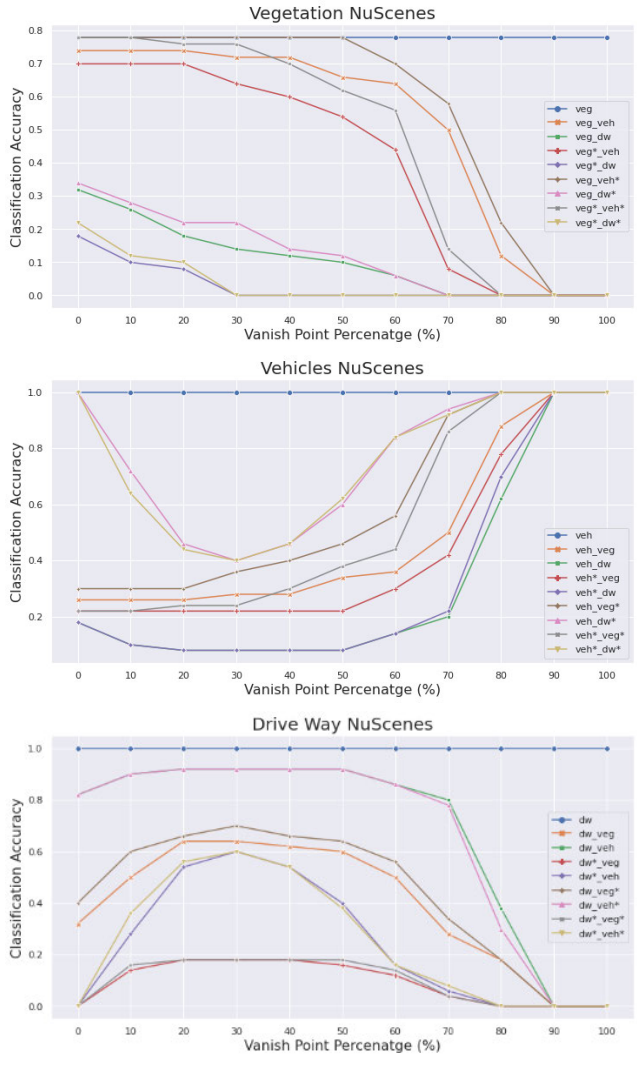


Fig. 8. Sensitivity to % LiDAR points vanished.

observed in Fig. 8. For Vegetation, we can observe that there is a slight decrease in classification accuracy till 60% vanishes points, then there is a steep fall in accuracy, and it converges to 0%. It can be observed for all the exchange combinations for Vegetation. For vehicles, we observe a different effect with respect to vanishing points. Initially, the accuracy is decreasing, and then it increases to the maximum. This can be explained as the Vehicle having a less boundary region (LiDAR points) so when there are fewer points our model is predicting the Vehicle class as a hard label. For Drive Way, the accuracy initially increases till 30% vanishes points and then significantly decreases to 0% when we vanish more points. Thus, we can conclude vanishing 50% of LiDAR data points is the optimal condition on fixed point clouds. There is further scope to extend our exchange vanishing attack to dynamic point clouds that consists of varying boundary regions for the same class. This analysis of dynamic point clouds would improve the performance of our vanishing attack, although exchanging of class labels is the main challenging issue.

VI. Conclusion

In this article, we proposed a novel vanishing attack using the exchange method on LiDAR point clouds in autonomous driving scenarios. The proposed LiDAR sensor attack is able to generate adversarial point clouds to effectively cause detection failure at the point level for targeted classes (Vegetation, Vehicle, and Driveway). Our proposed approach effectively vanishes or misclassifies more than 90% of targeted LiDAR point classes in the nuScenes and PandaSet datasets consisting of real-world scenarios. Moreover, we show that the targeted nature of our proposed attack helps to vanish objects by considering the semantic context in AVs, qualitatively and quantitatively. To further evaluate the robustness of our EVAA, adversarial defense methods can be introduced as an extension of our work.

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